LEARNING HUB MASTER SERIES • UNIVERSITY PHYSICS

University Physics Reference Handbook

Volume 9: Thermodynamics & Heat Transfer

A comprehensive, publication-grade academic treatise covering the Zeroth Law of Thermodynamics, thermal expansion and thermal stress, calorimetry and latent heat of phase transitions, Fourier's law of thermal conduction with series/parallel thermal resistance modeling, Newton's law of cooling for convection, Stefan-Boltzmann and Wien radiation laws, the First Law of Thermodynamics and path-dependent work in fundamental ideal gas expansions, the Second Law of Thermodynamics, heat engines and refrigerators, the Carnot cycle and maximum thermal efficiency, and the thermodynamic and statistical foundations of entropy.

SYNTHESIZED AUTHORITATIVE ACADEMIC SOURCES

- **University Lecture Slide Series:** *Heat Transfer (Conduction, Convection, Radiation)* • *Thermodynamics & Ideal Gas Laws*
- **Primary Textbook:** Ramamurti Shankar — *Fundamentals of Physics: Mechanics, Relativity, and Thermodynamics* (Yale University, Chapters 21–24)
- **Standard Formula Reference:** University Physics Formula Reference Sheet (embeds all fundamental physical equations directly)

Publication Standard: Pure English Master Edition • `Document/Physics/` Destination
Comprehensive Embedded Features: 4 High-Precision Scalable Vector SVG Visualizations • 5 Mental Calculation Fast-Tracks (⚡ Fast-Tracks) • 5 Step-by-Step Fully Solved Worked Problems • 3 Master Summary Appendices

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Part I: Thermal Foundations, Calorimetry & Heat Transfer Mechanisms

Chapter 1: Temperature, Thermal Equilibrium & Thermal Expansion

Thermodynamics is the macroscopic science governing energy transformations between heat, internal thermal motion, and mechanical work. While classical mechanics tracks individual point particles via Newton's laws, thermodynamics describes macroscopic ensembles consisting of ~10^{23} molecules using state variables such as pressure (P), volume (V), temperature (T), and internal energy (E_{int}).

1.1 Thermal Equilibrium & The Zeroth Law of Thermodynamics

When two bodies are placed in thermal contact (permitting thermal energy exchange through a diathermal boundary), heat flows spontaneously between them until macroscopic properties (pressure, density, electrical resistance) cease changing. At this stage, the two systems are in a state of **thermal equilibrium**.

▶ The Zeroth Law of Thermodynamics

If system A is in thermal equilibrium with system B, and system B is simultaneously in thermal equilibrium with system C, then system A is in thermal equilibrium with system C:

$$T_A = T_B \quad \text{and} \quad T_B = T_C \quad \Rightarrow \quad T_A = T_C$$

Physical Significance: The Zeroth Law rigorously establishes **temperature** as an objective, transitive scalar property inherent to any thermodynamic system. A third calibrated body (a thermometer) can verify whether two separate systems possess identical temperatures without placing them in direct contact.

1.2 Temperature Scales & Absolute Zero

The SI base unit of thermodynamic temperature is the **Kelvin** (K), defined with absolute zero (0 K) as the state where constituent molecules possess minimum possible quantum ground-state kinetic energy. The triple point of pure water is defined as 273.16 K (0.01°C).

$$T_K = T_C + 273.15, \qquad T_F = (9/5)\,T_C + 32, \qquad \Delta T_K = \Delta T_C = (5/9)\,\Delta T_F$$

1.3 Thermal Expansion in Solids and Liquids

As the temperature of a solid increases, the mean amplitude of atomic vibrations within asymmetric interatomic potential wells grows, increasing the average interatomic separation distance. Macroscopically, this manifests as dimensional expansion.

- **Linear Thermal Expansion:** For an initial length L_0 subject to temperature change ΔT:

$$\Delta L = \alpha\, L_0\, \Delta T \qquad \Rightarrow \qquad L(T) = L_0\,[1 + \alpha\, \Delta T]$$

  where α is the coefficient of linear expansion (in units of K^{-1} or $(°C)^{-1}$).
- **Areal (Surface) Thermal Expansion:** For an initial surface area $A_0 = L_1 L_2$:

$$A(T) = L_1(1 + \alpha\,\Delta T) \cdot L_2(1 + \alpha\,\Delta T) = A_0\,[1 + 2\alpha\,\Delta T + \alpha^2(\Delta T)^2]$$

Neglecting second-order terms $\alpha^2 \ll 1$:

$$\boldsymbol{\Delta A \approx 2\alpha\, A_0\, \Delta T \quad \Rightarrow \quad \gamma \approx 2\alpha}$$

- **Volumetric Thermal Expansion:** For isotropic materials with initial volume V_0:

$$\boldsymbol{\Delta V = \beta\, V_0\, \Delta T \quad \text{where} \quad \beta \approx 3\alpha}$$

- **Thermal Stress in Rigidly Constrained Solids:** If a structural beam of length L_0 and cross-sectional area A is rigidly clamped between immovable supports and heated by ΔT, it is prevented from expanding. The resulting thermal compressive stress σ and axial force F are given by Hooke's Law and Young's Modulus (E):

$$\boldsymbol{\sigma = F / A = -E\,\alpha\,\Delta T \quad \Rightarrow \quad F = -A\,E\,\alpha\,\Delta T}$$

TRICK TH-1 **Quick Volumetric Expansion & Thermal Stress Multipliers** **5 Seconds**

Standard Method (Slow): Cube $(1 + \alpha\,\Delta T)$ algebraically, expand binomial terms, subtract 1, and divide.
⚡ **Fast Hand Calculation Shortcut:** For isotropic solids, always use the **1-2-3 Dimensional Rule**:
• 1D Length: Fractional change = $\alpha\,\Delta T$ | • 2D Area: Fractional change = $2\alpha\,\Delta T$ | • 3D Volume: Fractional change = $3\alpha\,\Delta T$ ($\beta = 3\alpha$)
Constrained Thermal Stress: The mechanical stress is simply $\sigma = E \cdot (\Delta L/L_0) = E\,\alpha\,\Delta T$.

Example: A steel rail (E = 200 GPa, $\alpha = 12 \times 10^{-6}\ \text{K}^{-1}$) heated by ΔT = 40 K experiences compressive stress: $\sigma = (200 \times 10^9)(12 \times 10^{-6})(40)$ = **96.0 MPa** in under 5 seconds!

Chapter 2: Heat Capacity, Calorimetry & Latent Heat

2.1 Heat, Heat Capacity and Specific Heat

Heat (Q) is defined strictly as energy in transit across the boundary of a thermodynamic system as a consequence of a temperature gradient. Heat is never "contained" within a body; rather, once absorbed, it transforms into microscopic kinetic and potential energy (internal energy).

- **Heat Capacity (C):** The energy required to raise the temperature of an entire object by 1 K:

$$C = dQ / dT \qquad [\text{J/K}]$$

- **Specific Heat Capacity (c):** Heat capacity normalized per unit mass:

$$\boldsymbol{Q = m\,c\,\Delta T = m\,c\,(T_f - T_i) \qquad [\text{J/(kg·K)}]}$$

- **Molar Heat Capacity (C_m):** Heat capacity normalized per mole of substance:

$$Q = n\,C_m\,\Delta T \qquad [\text{J/(mol·K)}], \qquad \text{where } C_m = M\,c$$

2.2 The Principle of Calorimetry (Method of Mixtures)

For a collection of bodies enclosed within an adiabatic (perfectly insulated) container with no mechanical work performed:

$$\sum Q_i = 0 \quad \Leftrightarrow \quad Q_{\text{gained}} = -Q_{\text{lost}} = |Q_{\text{lost}}|$$

2.3 Phase Transformations & Latent Heat

When a pure substance undergoes a first-order phase transition (solid ↔ liquid, liquid ↔ vapor), the temperature remains strictly constant while thermal energy is absorbed or released to break or form intermolecular bonds:

$$Q = \pm m L$$

- **Latent Heat of Fusion (L_f):** Melting / Freezing transition. For water at 0°C: $L_f \approx 3.33 \times 10^5$ J/kg (333 kJ/kg).
- **Latent Heat of Vaporization (L_V):** Boiling / Condensation transition. For water at 100°C: $L_V \approx 2.26 \times 10^6$ J/kg (2260 kJ/kg).

Chapter 3: Heat Transfer Mechanisms: Conduction, Convection & Radiation

Heat spontaneously transfers from regions of higher temperature to regions of lower temperature via three distinct physical mechanisms: thermal conduction, fluid convection, and electromagnetic radiation.

3.1 Thermal Conduction & Fourier's Law

Conduction occurs via molecular and lattice vibrations (phonons) and free electron collisions without macroscopic bulk motion of matter. In one dimension, steady-state heat conduction is governed by **Fourier's Law**:

$$H = \frac{dQ}{dt} = -k A \frac{dT}{dx}$$

where H is the heat transfer rate in Watts (W = J/s), k is the thermal conductivity of the material [W/(m·K)], A is the cross-sectional area perpendicular to the heat flux, and dT/dx is the spatial temperature gradient. For a uniform slab of thickness L with surface temperatures T_H and T_C:

$$H = k A \frac{T_H - T_C}{L} = \frac{\Delta T}{R_{th}}$$

where $R_{th} = L / (k A)$ is defined as the **Thermal Resistance** (in K/W). This establishes an exact mathematical isomorphism with Ohm's law ($I = \Delta V / R$), where heat flow rate H plays the role of current, and temperature difference ΔT plays the role of potential difference.

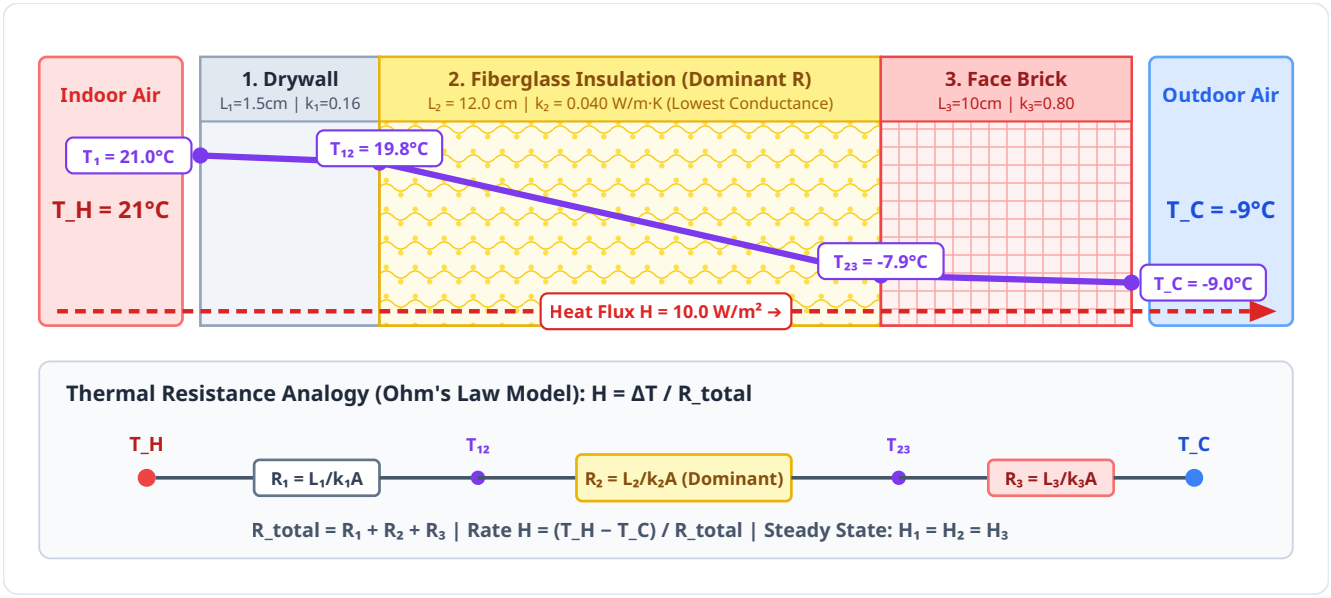


Figure 1: Steady-State Thermal Conduction Across a Multi-Layer Composite Wall. Heat flux H flows left-to-right from indoor air (T_H = 21°C) to outdoor air (T_C = −9°C) through drywall, fiberglass insulation, and exterior brick. The temperature profile $T(x)$ exhibits the steepest temperature drop across the high-resistance fiberglass layer. The electrical circuit analog directly illustrates series resistance addition: $R_{total} = \sum L_i / (k_i A)$.

TRICK TH-2 **2-Layer Interface Temperature & Conductance Weighting Rule** **10 Seconds**

Standard Method (Slow): Set $H_1 = k_1 A(T_H - T_{int})/L_1$ equal to $H_2 = k_2 A(T_{int} - T_C)/L_2$, cross multiply, expand parentheses, group terms, and solve for T_{int}.

⚡ **Fast Hand Calculation Shortcut:** Treat thermal conductances $G_i = k_i / L_i$ as weights! The interface temperature is the **conductance-weighted average**:

$$T_{int} = \frac{G_1 T_H + G_2 T_C}{G_1 + G_2}$$

Instant Ratio Rule: The temperature drop across any layer is directly proportional to its resistance: $\Delta T_i / \Delta T_{total} = R_i / R_{total}$. The layer with the highest R takes the lion's share of the temperature drop!

3.2 Thermal Convection

Convection is heat transfer accomplished through the macroscopic bulk motion of fluid packets (liquid or gas). It is categorized into *natural (free) convection* (driven by buoyancy differences from temperature-induced density gradients) and *forced convection* (driven by mechanical pumps, fans, or external winds). Macroscopic convective heat exchange is modeled empirically by **Newton's Law of Cooling**:

$$H_{conv} = h A (T_s - T_\infty)$$

where h is the convective heat transfer coefficient [W/(m^2·K)], A is the exposed surface area, T_s is the surface temperature, and T_∞ is the ambient fluid bulk temperature.

3.3 Thermal Radiation & The Stefan-Boltzmann Law

Thermal Radiation is electromagnetic energy emitted by all matter with a non-zero thermodynamic temperature ($T > 0$ K) due to rapid acceleration of thermal charge carriers and molecular dipole oscillations. Unlike conduction and convection, radiation requires no physical matter and propagates through a vacuum at the speed of light.

- **The Stefan-Boltzmann Law:** Total radiant power P emitted by a surface of area A at absolute temperature T:

$$P_{\text{emit}} = \varepsilon \, \sigma \, A \, T^4$$

where σ is the universal **Stefan-Boltzmann constant**:

$$\sigma = 5.670\,374 \times 10^{-8}\ \text{W} / (\text{m}^2 \cdot \text{K}^4)$$

and ε is the surface **emissivity** (dimensionless, $0 \le \varepsilon \le 1$). An ideal emitter is called a **blackbody** ($\varepsilon = 1$), absorbing and radiating with maximum theoretical efficiency.

- **Net Radiative Heat Transfer:** For an object at temperature T enclosed within surroundings at temperature T_{env}:

$$P_{\text{net}} = \varepsilon \, \sigma \, A \, (T^4 - T_{\text{env}}{}^4)$$

- **Wien's Displacement Law:** The peak wavelength λ_{max} of spectral emissive power is inversely proportional to temperature:

$$\lambda_{\text{max}} \, T = b \approx 2.898 \times 10^{-3}\ \text{m} \cdot \text{K}$$

Part II: The First Law of Thermodynamics & Ideal Gas Processes

Chapter 4: The First Law of Thermodynamics, Internal Energy & Sign Conventions

4.1 Work Done by an Expanding Gas

Consider a gas enclosed in a cylinder with a movable piston of cross-sectional area A. When the gas exerts pressure P and expands by an infinitesimal displacement dx, the mechanical work performed by the gas is:

$$dW = F\,dx = (P\,A)\,dx = P\,(A\,dx) = \boldsymbol{P\,dV}$$

For a finite expansion from initial volume V_i to final volume V_f along a continuous quasi-static path:

$$\boldsymbol{W = \int_{V_i}^{V_f} P\,dV} \qquad \textbf{[Joules]}$$

Geometric Meaning: Work is the integral area underneath the trajectory curve in the P-V plane. Crucially, because the path between state (P_i, V_i) and state (P_f, V_f) can vary, **work is path-dependent** (an inexact differential δW). Work is NOT a state property of a system!

4.2 The First Law of Thermodynamics

The First Law expresses universal conservation of energy for a thermodynamic system:

▶ The First Law of Thermodynamics

$$\boldsymbol{\Delta E_{int} = Q - W} \qquad \Leftrightarrow \qquad \boldsymbol{dE_{int} = dQ - dW}$$

where:

- **ΔE_{int}:** Change in internal energy. Internal energy is a **state function** (depends only on the current state coordinates T, P, V, and $\oint dE_{int} = 0$ for any closed cycle).
- **Q:** Heat added to the system ($Q > 0$ when absorbed, $Q < 0$ when rejected).
- **W:** Work done *by* the system ($W > 0$ for expansion $dV > 0$, $W < 0$ for compression $dV < 0$).

Alternative Sign Convention: In chemistry and mechanics textbooks writing $\Delta E_{int} = Q + W_{on}$, $W_{on} = -\int P\,dV$ represents work done *on* the gas. Both forms are physically identical.

4.3 Microscopic Equipartition & Internal Energy of an Ideal Gas

According to the classical **Equipartition Theorem**, each quadratic degree of freedom of a particle in thermal equilibrium contributes an average energy of (1/2) $k_B\,T$ per molecule, or (1/2) $R\,T$ per mole.

- **Monatomic Gas (He, Ne, Ar):** 3 translational degrees of freedom ($f = 3$):

$$E_{\text{int}} = (3/2)\, n\, R\, T \quad \Rightarrow \quad C_{\text{v}} = (3/2)\, R, \quad C_{\text{p}} = (5/2)\, R, \quad \gamma = 5/3 \approx 1.67$$

- **Diatomic Gas at Room Temperature (N_2, O_2, Air):** 3 translational + 2 rotational DOF ($f = 5$):

$$E_{\text{int}} = (5/2)\, n\, R\, T \quad \Rightarrow \quad C_{\text{v}} = (5/2)\, R, \quad C_{\text{p}} = (7/2)\, R, \quad \gamma = 7/5 = 1.40$$

Crucial Insight (Joule's Free Expansion): For an ideal gas, internal energy is strictly a function of temperature alone: $\boldsymbol{\Delta E_{\text{int}} = n\, C_{\text{v}}\, \Delta T}$ regardless of the thermodynamic path taken!

Chapter 5: The Four Fundamental Ideal Gas Processes

5.1 Detailed Mathematical Breakdown of the Four Processes

1. **Isobaric Process (Constant Pressure, P = const):**
   - *Work:* $W = \int P\, dV = P\,(V_{\text{f}} - V_{\text{i}}) = P\, \Delta V = n\, R\, \Delta T$.
   - *Internal Energy:* $\Delta E_{\text{int}} = n\, C_{\text{v}}\, \Delta T$.
   - *Heat Added:* $Q = \Delta E_{\text{int}} + W = n\,(C_{\text{v}} + R)\, \Delta T = \boldsymbol{n\, C_{\text{p}}\, \Delta T}$.
   - **Mayer's Relation:** $C_{\text{p}} - C_{\text{v}} = R$. Ratio of specific heats: $\gamma = C_{\text{p}} / C_{\text{v}}$.
2. **Isochoric Process (Constant Volume, V = const):**
   - *Work:* Since $dV = 0$, $\boldsymbol{W = 0}$.
   - *First Law:* $\boldsymbol{Q = \Delta E_{\text{int}} = n\, C_{\text{v}}\, \Delta T} = \dfrac{V\, \Delta P}{\gamma - 1}$.
3. **Isothermal Process (Constant Temperature, T = const):**
   - *Internal Energy:* Since $\Delta T = 0$, $\boldsymbol{\Delta E_{\text{int}} = 0}$.
   - *Work:* Substituting $P = n\, R\, T / V$:

$$\boldsymbol{W = \int_{V_{\text{i}}}^{V_{\text{f}}} \frac{n\, R\, T}{V}\, dV = n\, R\, T \ln(V_{\text{f}} / V_{\text{i}}) = n\, R\, T \ln(P_{\text{i}} / P_{\text{f}})}$$

   - *Heat:* By the First Law, $\boldsymbol{Q = W = n\, R\, T \ln(V_{\text{f}} / V_{\text{i}})}$.
4. **Adiabatic Process (Zero Heat Exchange, Q = 0):**
   An adiabatic process occurs so rapidly or in such well-insulated containers that no heat transfers: $\boldsymbol{Q = 0}$.
   - *First Law:* $dE_{\text{int}} = -dW \Rightarrow n\, C_{\text{v}}\, dT = -P\, dV = -(n\, R\, T / V)\, dV$.

   Dividing by T and using $R/C_{\text{v}} = \gamma - 1$:

$$\frac{dT}{T} + (\gamma - 1)\frac{dV}{V} = 0 \quad \Rightarrow \quad \boldsymbol{T\, V^{\gamma - 1} = \text{const}}$$

   Combining with the ideal gas law $P\, V = n\, R\, T$ yields the three equivalent **Adiabatic Relations**:

$$\boldsymbol{P\, V^{\gamma} = \text{const}, \quad T\, V^{\gamma - 1} = \text{const}, \quad T^{\gamma}\, P^{1 - \gamma} = \text{const}}$$

   - *Work Done:* $W = -\Delta E_{\text{int}} = -n\, C_{\text{v}}\,(T_{\text{f}} - T_{\text{i}}) = \dfrac{P_{\text{i}}\, V_{\text{i}} - P_{\text{f}}\, V_{\text{f}}}{\gamma - 1} = \dfrac{n\, R\,(T_{\text{i}} - T_{\text{f}})}{\gamma - 1}$.

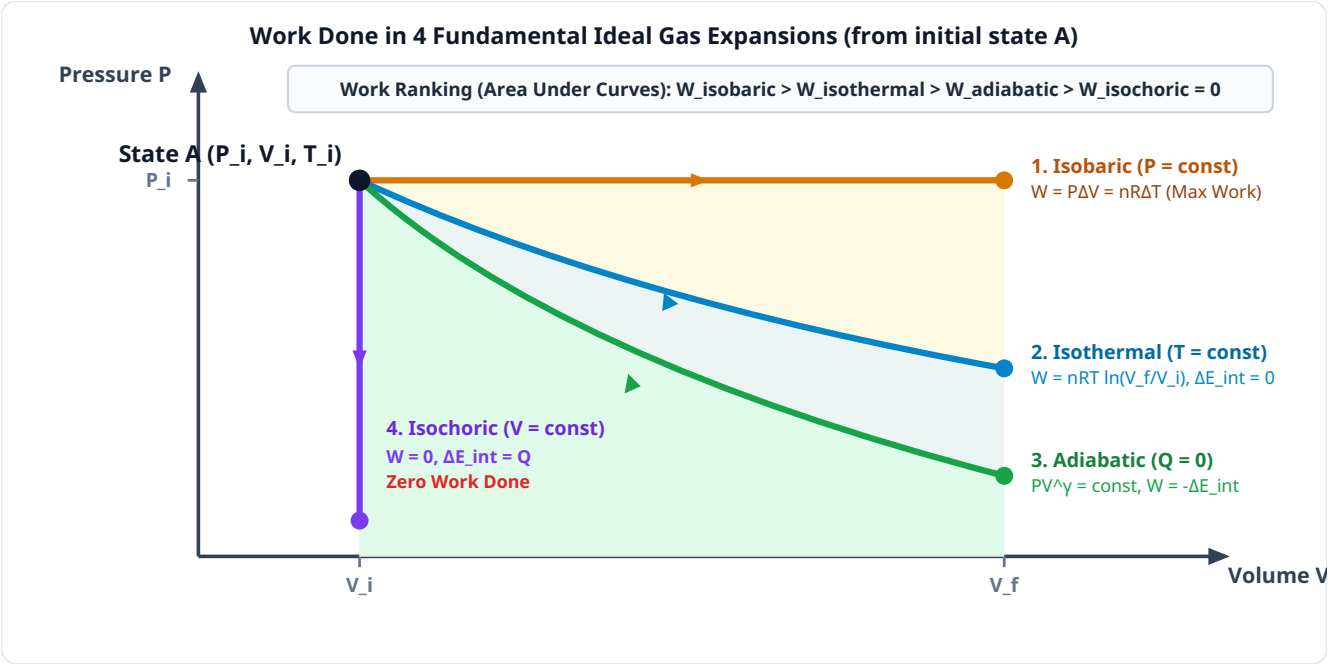


Figure 2: P-V Curves and Work Areas for the Four Fundamental Ideal Gas Expansions. Originating from initial state $A(P_i, V_i)$, expansion to final volume V_f produces work equal to the area under each curve. The isobaric path sustains maximum pressure, yielding maximum work. The isothermal curve follows $P \propto V^{-1}$, while the adiabatic curve is steeper ($P \propto V^{-\gamma}$ with $\gamma > 1$) because temperature drops as work is extracted from internal energy. The isochoric path undergoes zero volume change, performing zero work ($W = 0$).

TRICK TH-3 **Isobaric Energy Partitioning & Adiabatic Exponent Shortcut** **5 Seconds**

⚡ **Fast Hand Calculation Shortcut:** In any isobaric expansion of an ideal gas, the fraction of added heat converted into work and internal energy is fixed entirely by γ:

$$\frac{W}{Q} = \frac{R}{C_p} = \frac{\gamma - 1}{\gamma} \quad \text{and} \quad \frac{\Delta E_{int}}{Q} = \frac{C_v}{C_p} = \frac{1}{\gamma}$$

• **Monatomic Gas (γ = 5/3):** Exactly **40%** of heat becomes work ($W = 0.40\,Q$), and **60%** raises internal energy ($\Delta E_{int} = 0.60\,Q$).

• **Diatomic Gas (γ = 7/5):** Exactly **28.6%** (2/7) of heat becomes work ($W = (2/7)Q$), and **71.4%** (5/7) raises internal energy ($\Delta E_{int} = (5/7)Q$).

Adiabatic Temperature Ratio: For compression ratio $r = V_1/V_2$, $\boldsymbol{T_2 = T_1 \cdot r^{\gamma-1}}$. For diatomic gas ($\gamma - 1 = 0.4 = 2/5$), evaluate $r^{0.4}$ instantly!

Part III: The Second Law, Heat Engines, Carnot Cycle & Entropy

Chapter 6: Heat Engines, Refrigerators & The Second Law of Thermodynamics

6.1 Cyclic Thermodynamic Processes

A thermodynamic cycle is a sequence of processes that returns the working fluid to its exact initial state: (P_1, V_1, T_1). Because internal energy E_{int} is a state function:

$$\Delta E_{\text{int, cycle}} = \oint dE_{\text{int}} = 0 \quad \Rightarrow \quad W_{\text{net}} = Q_{\text{net}} = Q_{\text{H}} - |Q_{\text{C}}|$$

where W_{net} is the geometric area enclosed by the closed trajectory loop in the *P-V* diagram. A clockwise cycle encloses positive work ($W_{net} > 0$, heat engine), while a counter-clockwise cycle encloses negative work ($W_{net} < 0$, requiring work input, refrigerator or heat pump).

6.2 Thermal Efficiency of Heat Engines

A heat engine operates between a hot thermal reservoir at temperature T_H and a cold reservoir at temperature T_C. It absorbs heat $Q_H > 0$, performs useful mechanical work W_{net}, and exhausts waste heat $|Q_C|$ to the cold sink. The **Thermal Efficiency (η)** is defined as the ratio of useful work output to the paid energy input:

$$\eta = \frac{W_{\text{net}}}{Q_{\text{H}}} = \frac{Q_{\text{H}} - |Q_{\text{C}}|}{Q_{\text{H}}} = 1 - \frac{|Q_{\text{C}}|}{Q_{\text{H}}}$$

6.3 Refrigerators and Heat Pumps (Reversed Cycles)

A refrigerator or heat pump transfers heat from a lower temperature reservoir to a higher temperature reservoir by consuming external work input W_{in}.

- **Refrigerator Coefficient of Performance (COP_{ref}):** The objective is cooling the cold space:

$$\text{COP}_{\text{ref}} = K = \frac{|Q_{\text{C}}|}{W_{\text{in}}} = \frac{|Q_{\text{C}}|}{Q_{\text{H}} - |Q_{\text{C}}|}$$

- **Heat Pump Coefficient of Performance (COP_{hp}):** The objective is heating the hot space:

$$\text{COP}_{\text{hp}} = \frac{Q_{\text{H}}}{W_{\text{in}}} = \frac{|Q_{\text{C}}| + W_{\text{in}}}{W_{\text{in}}} = \text{COP}_{\text{ref}} + 1$$

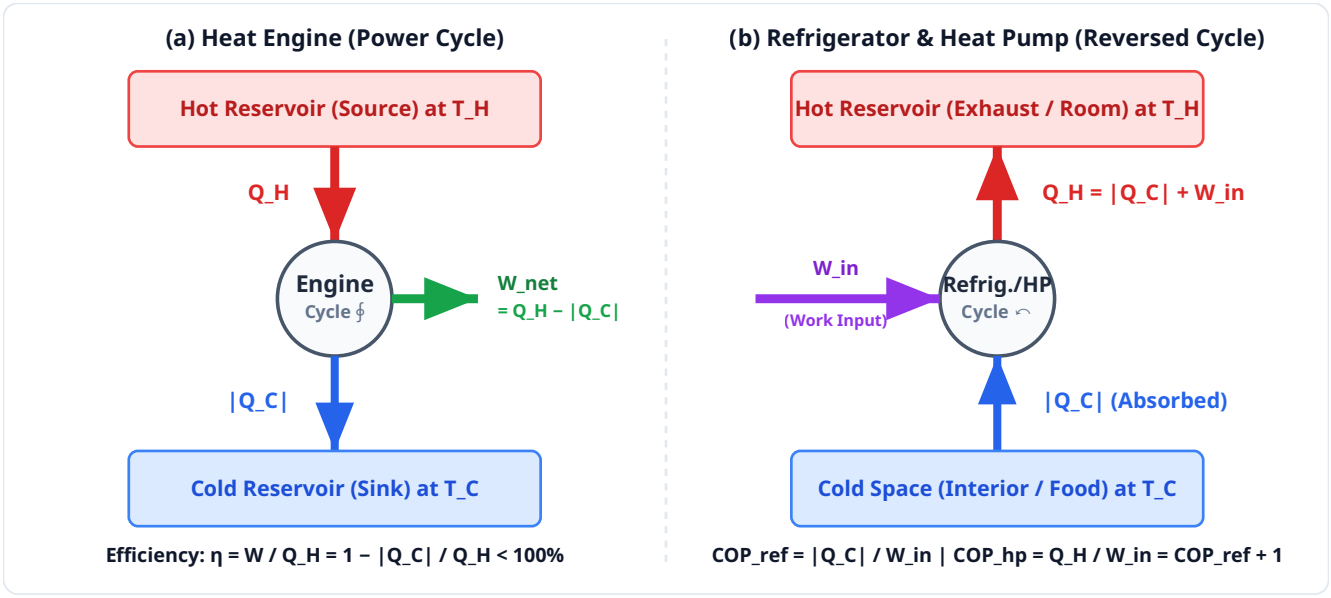


Figure 3: Energy Flow Sankey Diagrams for Heat Engines and Refrigerators / Heat Pumps. (a) A heat engine absorbs heat Q_H from the hot source, outputs useful work W_{net}, and inevitably discharges waste heat $|Q_C|$ to the cold sink. (b) In a refrigerator / heat pump, work W_{in} drives heat extraction $|Q_C|$ from the cold interior and exhausts $Q_H = |Q_C| + W_{in}$ to the hot environment.

6.4 Classical Statements of the Second Law of Thermodynamics

▶ Fundamental Statements of the Second Law

- **Kelvin-Planck Statement (Engine Formulation):**
  It is impossible to construct a thermodynamic device operating in a cycle that produces no effect other than the extraction of heat from a single thermal reservoir and the conversion of all that heat into an equivalent amount of mechanical work. (No heat engine can achieve η = 100%; waste heat rejection $|Q_C| > 0$ is physically mandatory).
- **Clausius Statement (Refrigerator Formulation):**
  It is impossible to construct a thermodynamic device operating in a cycle whose sole effect is the transfer of heat from a cooler body to a warmer body without external work input. (Heat cannot flow spontaneously from cold to hot; refrigeration strictly requires work input $W_{in} > 0$).

Theorem of Equivalence: If either statement were false, a composite engine could be constructed that violates the other. Thus, both statements represent identical physical truth.

Chapter 7: The Carnot Cycle & Maximum Theoretical Limit

In 1824, French engineer Sadi Carnot proved that any heat engine operating between two fixed thermal reservoirs at temperatures T_H and T_C is bounded by a maximum possible theoretical efficiency achieved only by a reversible cycle.

7.1 The Four Stages of the Carnot Cycle

1. **Stage A → B: Reversible Isothermal Expansion at T_H.** Working substance is in thermal contact with hot reservoir at T_H. Absorbs heat $Q_H = n\,R\,T_H \ln(V_B/V_A)$.

2. **Stage B → C: Reversible Adiabatic Expansion.** Cylinder thermally isolated. Gas expands, doing work against piston. Temperature decreases from T_H to T_C according to $T_H V_B^{\gamma-1} = T_C V_C^{\gamma-1}$.
3. **Stage C → D: Reversible Isothermal Compression at T_C.** Working substance in thermal contact with cold sink at T_C. Work done on gas; rejects heat $|Q_C| = n R T_C \ln(V_C/V_D)$.
4. **Stage D → A: Reversible Adiabatic Compression.** Cylinder thermally isolated. Work done on gas raises temperature from T_C back to T_H according to $T_C V_D^{\gamma-1} = T_H V_A^{\gamma-1}$.

Dividing the two adiabatic relations:

$$\frac{V_B}{V_A} = \frac{V_C}{V_D} \quad \Rightarrow \quad \frac{|Q_C|}{Q_H} = \frac{T_C}{T_H}$$

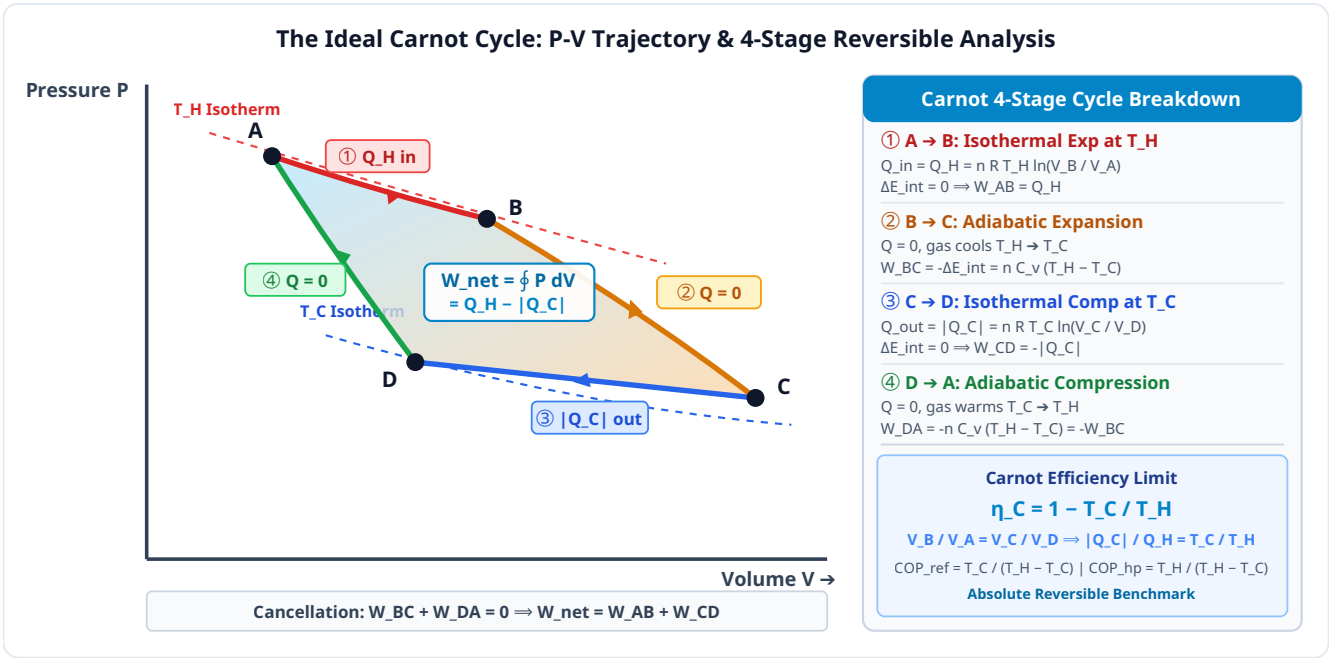


Figure 4: The Carnot Cycle on P-V Coordinates. Consisting of two reversible isotherms (T_H and T_C) connected by two reversible adiabats ($Q = 0$), the shaded central region represents net work delivered: $W_{net} = \oint P\,dV = Q_H - |Q_C|$. The cycle achieves the absolute maximum thermal efficiency possible between temperatures T_H and T_C.

▶ Carnot's Efficiency & COP Limits

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H}$$

$$\text{COP}_{\text{Carnot, ref}} = \frac{T_C}{T_H - T_C} \quad \text{and} \quad \text{COP}_{\text{Carnot, hp}} = \frac{T_H}{T_H - T_C}$$

Carnot's Theorem: No real engine operating between two specified heat reservoirs can have an efficiency higher than a Carnot engine operating between the same two reservoirs: $\eta_{\text{real}} \leq \eta_{\text{Carnot}}$. All reversible engines have exactly $\eta = \eta_{\text{Carnot}}$.

TRICK TH-4 **Instant Carnot Efficiency & COP Cross-Conversion** **3 Seconds**

Standard Method (Slow): Convert both temperatures to Kelvin, substitute into $1 - T_C/T_H$, find a common denominator, compute decimals, and invert for COP.

⚡ **Fast Hand Calculation Shortcut:** Always write efficiency as the **temperature drop ratio**:

$$\eta_C = \frac{\Delta T}{T_H} \quad \text{and} \quad \text{COP}_{\text{Carnot}} = \frac{T_C}{\Delta T} = \frac{1 - \eta_C}{\eta_C}$$

Example: An engine operating at 25% Carnot efficiency ($\eta_C = 0.25$) when reversed into a refrigerator has: COP = (1 − 0.25) / 0.25 = 0.75 / 0.25 = **3.0** in 3 seconds flat!

Chapter 8: Entropy & The Principle of Entropy Increase

8.1 The Thermodynamic Definition of Entropy

From the Carnot relationship $|Q_C| / Q_H = T_C / T_H$, treating heat algebraically ($Q_H > 0$ into engine, $Q_C < 0$ out):

$$\frac{Q_H}{T_H} + \frac{Q_C}{T_C} = 0 \quad \Rightarrow \quad \oint \frac{dQ_{\text{rev}}}{T} = 0$$

Because the cyclic integral of dQ_{rev}/T vanishes along any arbitrary reversible loop, the quantity is a **state function**. This defines the thermodynamic property **Entropy (S)**:

$$dS = \frac{dQ_{\text{rev}}}{T} \quad \Rightarrow \quad \Delta S = S_B - S_A = \int_A^B \frac{dQ_{\text{rev}}}{T} \quad [\text{J/K}]$$

8.2 Clausius Inequality & The Principle of Entropy Increase

For any real cyclic process (reversible or irreversible), the **Clausius Inequality** states:

$$\oint \frac{dQ}{T} \leq 0 \qquad \textbf{(equality holds strictly for reversible cycles)}$$

When applied to the isolated universe (system + surroundings):

▶ The Principle of Entropy Increase (The Second Law)

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$$

The total entropy of an isolated system never decreases. In all natural, spontaneous (irreversible) physical processes, the entropy of the universe strictly increases ($\Delta S_{\text{univ}} > 0$). Entropy remains constant ($\Delta S_{\text{univ}} = 0$) only during idealized, quasi-static reversible transformations.

8.3 Calculating Entropy Changes for Ideal Gases

From the First Law, $dQ_{\text{rev}} = dE_{\text{int}} + P\,dV = n\,C_{\text{v}}\,dT + (n\,R\,T / V)\,dV$. Dividing by T:

$$dS = n\,C_{\text{v}}\frac{dT}{T} + n\,R\frac{dV}{V}$$

Integrating from initial state (T_{i}, V_{i}) to final state (T_{f}, V_{f}):

$$\Delta S = n\,C_{\text{v}}\ln(T_{\text{f}} / T_{\text{i}}) + n\,R\ln(V_{\text{f}} / V_{\text{i}}) = n\,C_{\text{p}}\ln(T_{\text{f}} / T_{\text{i}}) - n\,R\ln(P_{\text{f}} / P_{\text{i}})$$

- **Isothermal Expansion (T = const):** $\Delta S = n\,R\ln(V_{\text{f}} / V_{\text{i}}) = Q / T$.
- **Isochoric Heating (V = const):** $\Delta S = n\,C_{\text{v}}\ln(T_{\text{f}} / T_{\text{i}})$.
- **Isobaric Expansion (P = const):** $\Delta S = n\,C_{\text{p}}\ln(T_{\text{f}} / T_{\text{i}})$.
- **Reversible Adiabatic (Isentropic):** $dQ_{\text{rev}} = 0 \Rightarrow \Delta S = 0$.
- **Phase Change at Constant T_{trans}: $\Delta S = \pm\, m\,L / T_{\text{trans}}$.**

TRICK TH-5 **Free Expansion & Isothermal Volume Doubling Entropy Shortcut** **5 Seconds**

The Free Expansion Paradox: When an ideal gas undergoes free expansion into a vacuum chamber (Joule expansion), no work is done ($W = 0$) and no heat is exchanged ($Q = 0$), so $\Delta E_{\text{int}} = 0$ and $\Delta T = 0$. Although $Q = 0$, the process is completely **irreversible**, so $\Delta S \neq Q/T$!

⚡ **Fast Hand Calculation Shortcut:** Because entropy is a state function, evaluate ΔS along a reversible isotherm between the same endpoints:

$$\Delta S_{\text{gas}} = n\,R\ln(V_{\text{f}} / V_{\text{i}})$$

- **Volume Doubling ($V_{\text{f}} = 2V_{\text{i}}$):** $\Delta S = n\,R\ln 2 \approx \mathbf{0.693\,n\,R}$.
- **Volume Tripling ($V_{\text{f}} = 3V_{\text{i}}$):** $\Delta S = n\,R\ln 3 \approx \mathbf{1.099\,n\,R}$.

Because surroundings undergo zero change, $\Delta S_{\text{universe}} = \Delta S_{\text{gas}} > 0$, confirming irreversible entropy generation!

8.4 Microscopic / Statistical Interpretation of Entropy

Ludwig Boltzmann united macroscopic thermodynamics with microscopic statistical mechanics through the famous relation:

$$S = k_B \ln \Omega$$

where k_B = 1.380 649 × 10^{-23} J/K is Boltzmann's constant, and Ω is the thermodynamic probability (the total number of accessible microscopic quantum states compatible with the specified macroscopic state variables E, V, N). Entropy is therefore a direct physical measure of molecular configuration disorder and statistical dispersion.

Part IV: Step-by-Step Rigorous Worked Problems

Problem 1: Multi-Layer Insulated Wall Heat Loss & Interface Temperature

Chapter 3 • Fast-Track TH-2

Problem Statement: An exterior building wall of total surface area A = 24.0 m^2 separates indoor heated air at T_H = 21.0°C from outdoor cold air at T_C = −9.0°C. The wall consists of three contiguous layers in series: (1) an inner drywall sheet of thickness L_1 = 1.50 cm (0.0150 m) with thermal conductivity k_1 = 0.160 W/(m·K); (2) a middle fiberglass insulation batting of thickness L_2 = 12.0 cm (0.120 m) with k_2 = 0.0400 W/(m·K); and (3) an outer face brick layer of thickness L_3 = 10.0 cm (0.100 m) with k_3 = 0.800 W/(m·K). Assuming steady-state 1D heat flow, determine: (a) The total thermal resistance $R_{th,\,total}$ of the composite wall; (b) The rate of heat loss H through the wall in Watts; (c) The interface temperatures T_{12} (drywall-fiberglass) and T_{23} (fiberglass-brick); (d) The total energy lost in kilowatt-hours (kWh) over a 24.0-hour period, and compare with an uninsulated wall (brick + drywall only).

Step 1: Given Parameters & Physical Setup

A = 24.0 m^2, $\Delta T = T_H - T_C = 21.0 - (-9.0) = 30.0$ K.

Layer 1 (Drywall): L_1 = 0.0150 m, k_1 = 0.160 W/(m·K).

Layer 2 (Fiberglass): L_2 = 0.120 m, k_2 = 0.0400 W/(m·K).

Layer 3 (Brick): L_3 = 0.100 m, k_3 = 0.800 W/(m·K).

Step 2: Symbolic Algebraic Derivations

Thermal resistance for layer i: $R_i = L_i / (k_i A)$. Total resistance: $R_{total} = R_1 + R_2 + R_3$.

Heat loss rate: $H = \Delta T / R_{total}$.

Interface temperatures: $T_{12} = T_H - H R_1$ and $T_{23} = T_{12} - H R_2 = T_C + H R_3$.

Daily energy loss: $Q_{24h} = H \cdot \Delta t$, converted via 1 kWh = 3.60 × 10^6 J.

Step 3: Numerical Evaluations with SI Units

• R_1 = 0.0150 / (0.160 × 24.0) = 0.0150 / 3.84 = **3.906 × 10^{-3} K/W**.

• R_2 = 0.120 / (0.0400 × 24.0) = 0.120 / 0.960 = **1.250 × 10^{-1} K/W = 0.1250 K/W**.

• R_3 = 0.100 / (0.800 × 24.0) = 0.100 / 19.2 = **5.208 × 10^{-3} K/W**.

(a) $\boldsymbol{R_{total}}$ **= 0.003906 + 0.1250 + 0.005208 = 0.13411 K/W**.

(b) $\boldsymbol{H}$ **= 30.0 K / 0.13411 K/W = 223.69 W ≈ 224 W**.

(c) Interface temperatures:

• $\Delta T_1 = H R_1$ = 223.69 × 0.003906 = 0.87 K ⇒ $\boldsymbol{T_{12}}$ **= 21.0 − 0.87 = 20.13°C**.

• $\Delta T_2 = H\,R_2 = 223.69 \times 0.1250 = 27.96$ K ⇒ $\mathbf{T_{23} = 20.13 - 27.96 = -7.83°C}$.

• Check layer 3: $T_{23} - T_C = -7.83 - (-9.00) = 1.17$ K = $223.69 \times 0.005208 = 1.17$ K ✓.

(d) Energy loss over 24 hours (Δt = 86,400 s):

$Q_{24h} = 223.69$ W × 86,400 s = 1.9327×10^7 J = **5.37 kWh**.

Comparison without insulation (Fiberglass removed): $R_{uninsulated} = R_1 + R_3 = 0.009114$ K/W.

$H_{uninsulated} = 30.0 / 0.009114 = 3291.6$ W ⇒ $Q_{uninsulated}$ = **79.0 kWh/day**.

The insulation reduces heat loss and heating costs by **93.2%**!

> **Physical Sanity Check:** Notice that the fiberglass represents $R_2/R_{total} = 0.125 / 0.1341 = 93.2\%$ of the total thermal resistance. In accordance with Fast-Track TH-2, it carries exactly 27.96 K / 30.0 K = 93.2% of the total temperature drop! The drywall surface stays comfortably warm at 20.1°C, preventing indoor moisture condensation.

Problem 2: Complete 3-Stage Ideal Gas Thermodynamic Cycle Analysis

Chapter 5 • Fast-Track TH-3

Problem Statement: An engine operates on n = 1.50 moles of an ideal monatomic gas ($C_v = (3/2)R$, $C_p = (5/2)R$, γ = 5/3, R = 8.314 J/(mol·K)) through a closed 3-stage cycle: • **State 1 → 2 (Isobaric Expansion):** From P_1 = 200 kPa, V_1 = 0.0200 m^3 to V_2 = 0.0500 m^3 at constant pressure P_2 = 200 kPa. • **State 2 → 3 (Isochoric Cooling):** Volume held constant at V_3 = 0.0500 m^3 while pressure drops to P_3 = 80.0 kPa. • **State 3 → 1 (Linear Compression):** Gas returns directly to state 1 along a straight-line path in the P-V diagram. For each stage and for the complete cycle, calculate: (a) Temperatures at all three states (T_1, T_2, T_3); (b) Work done W, heat absorbed/rejected Q, and internal energy change ΔE_{int}; (c) Net cycle work W_{net}, total heat input Q_{in}, and overall thermal efficiency η.

Step 1: State Temperatures via Ideal Gas Law ($T = P\,V / n\,R$)

• $T_1 = (200 \times 10^3 \times 0.0200) / (1.50 \times 8.314) = 4000 / 12.471 =$ **320.74 K**.

• $T_2 = (200 \times 10^3 \times 0.0500) / 12.471 = 10000 / 12.471 =$ **801.86 K**.

• $T_3 = (80 \times 10^3 \times 0.0500) / 12.471 = 4000 / 12.471 =$ **320.74 K**. (Notice $T_3 = T_1$!).

Step 2: Stage-by-Stage Thermodynamic Calculations

Stage 1 → 2 (Isobaric, P = 200 kPa):

• $W_{12} = P\,(V_2 - V_1) = 200 \times 10^3 \times (0.0500 - 0.0200) =$ **+6,000 J**.

• $\Delta E_{int,\,12} = n\,C_v\,(T_2 - T_1) = 1.50 \times (1.5 \times 8.314) \times (801.86 - 320.74) = (3/2) \times (10000 - 4000) =$ **+9,000 J**.

• $Q_{12} = \Delta E_{int} + W = 9000 + 6000 =$ **+15,000 J**. (Matches Fast-Track TH-3: $W = 0.40Q$, $\Delta E = 0.60Q$!).

Stage 2 → 3 (Isochoric, V = 0.0500 m^3):

• $W_{23} =$ **0 J** (rigid volume).

• $\Delta E_{int,\,23} = n\,C_v\,(T_3 - T_2) = (3/2)\,V_2\,(P_3 - P_2) = 1.5 \times 0.0500 \times (80 - 200) \times 10^3 =$ **−9,000 J**.

• $Q_{23} = \Delta E_{int,\,23} =$ **−9,000 J** (heat rejected to cold sink).

Stage 3 → 1 (Linear Path on P-V Diagram):

• W_{31} = Area under straight line (trapezoid): $W_{31} = \frac{P_3 + P_1}{2}(V_1 - V_3) = \frac{(80 + 200) \times 10^3}{2}(0.0200 - 0.0500) = 140 \times 10^3 \times (-0.0300) =$ **−4,200 J**.

• $\Delta E_{\text{int, 31}}$: Since $T_3 = T_1 = 320.74$ K, $\mathbf{\Delta E_{int,\ 31} = 0\ J}$.
• $Q_{31} = \Delta E_{\text{int, 31}} + W_{31} = 0 + (-4200) = \mathbf{-4{,}200\ J}$ (heat rejected).

Step 3: Complete Cycle Net Quantities & Thermal Efficiency

• **Net Work:** $W_{net} = W_{12} + W_{23} + W_{31} = 6000 + 0 - 4200 = \mathbf{+1{,}800\ J}$.
Geometric Check: Triangle area in *P-V* plane = (1/2) × base × height = (1/2) × (0.0500 − 0.0200) × (200 − 80) × 10^3 = 0.5 × 0.0300 × 120,000 = **1,800 J** ✓.
• **Total Internal Energy Change:** $\Delta E_{\text{int, net}} = 9000 - 9000 + 0 = \mathbf{0\ J}$ ✓ (State function requirement satisfied).
• **Heat Absorbed:** $Q_{in} = Q_{12} = \mathbf{15{,}000\ J}$. Heat rejected $|Q_{out}| = |Q_{23} + Q_{31}| = 9000 + 4200 = 13{,}200$ J.
• $Q_{net} = 15{,}000 - 13{,}200 = 1{,}800\ \text{J} = W_{net}$ ✓.
• **Thermal Efficiency:**

$$\eta = \frac{W_{\text{net}}}{Q_{\text{in}}} = \frac{1{,}800\ \text{J}}{15{,}000\ \text{J}} = 0.120 = 12.0\%$$

Physical Sanity Check: Carnot efficiency between extreme cycle temperatures ($T_H = 801.86$ K, $T_C = 320.74$ K) is $\eta_{Carnot} = 1 - (320.74 / 801.86) = 60.0\%$. Because our triangular cycle is non-ideal and includes dissipative heat rejection during linear compression, $\eta = 12.0\% < \eta_{Carnot} = 60.0\%$, rigorously obeying the Second Law of Thermodynamics!

Problem 3: Adiabatic Compression & Temperature Rise in a Diesel Engine

Chapter 5 • Fast-Track TH-3

Problem Statement: In a heavy-duty diesel engine cylinder, ambient intake air (treated as an ideal diatomic gas with γ = 1.400, $C_v = (5/2)R = 20.785$ J/(mol·K), and molar mass $M = 28.97$ g/mol) enters at initial state $P_1 = 100$ kPa (1.00 bar), $T_1 = 300$ K (27°C), and initial cylinder volume $V_1 = 900\ \text{cm}^3 = 9.00 \times 10^{-4}\ \text{m}^3$. The piston rapidly compresses the gas quasi-statically and adiabatically with a volumetric compression ratio $r = V_1 / V_2 = 18.0$. Determine: (a) The final compressed volume V_2 and final peak pressure P_2; (b) The final gas temperature T_2 in Kelvin and °C, verifying that it exceeds the auto-ignition threshold of atomized diesel fuel (≈ 550 K); (c) The mechanical work performed on the gas during compression W_{comp}; (d) The change in internal energy ΔE_{int}.

Step 1: Given Parameters & Number of Moles

$P_1 = 1.00 \times 10^5$ Pa, $T_1 = 300$ K, $V_1 = 9.00 \times 10^{-4}\ \text{m}^3$, $r = 18.0$, γ = 1.40.
$n = P_1 V_1 / (R T_1) = (1.00 \times 10^5 \times 9.00 \times 10^{-4}) / (8.314 \times 300) = 90.0 / 2494.2 = \mathbf{0.036083\ mol}$.

Step 2: Symbolic Derivations via Adiabatic Relations

(a) $V_2 = V_1 / r$.
From $P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$: $\boldsymbol{P_2 = P_1 \cdot r^{\gamma}}$.
(b) From $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$: $\boldsymbol{T_2 = T_1 \cdot r^{\gamma-1}}$.
(c) Adiabatic work: $W = -\Delta E_{int} = (P_1 V_1 - P_2 V_2) / (\gamma - 1)$. Work done *on* the gas: $W_{on} = -W = \Delta E_{int} = n C_v (T_2 - T_1)$.

Step 3: Numerical Evaluations

• $V_2 = 9.00 \times 10^{-4}\ m^3 / 18.0 =$ **$5.00 \times 10^{-5}\ m^3$ (50.0 cm^3)**.

• $r^{\gamma} = 18.0^{1.40} = 57.198 \Rightarrow$ **$P_2 = 100\ kPa \times 57.198 = 5{,}720\ kPa = 5.72\ MPa$ (57.2 atm)**.

• $r^{\gamma-1} = 18.0^{0.40} = 3.1777 \Rightarrow$ **$T_2 = 300\ K \times 3.1777 = 953.3\ K \approx 680°C$**.

• Temperature rise: $\Delta T = 953.3 - 300.0 =$ **+653.3 K**.

• Internal energy change: $\Delta E_{int} = n\ C_v\ \Delta T = 0.036083\ mol \times (2.5 \times 8.314\ J/(mol\cdot K)) \times 653.3\ K = 0.036083 \times 20.785 \times 653.3 =$ **+490.0 J**.

• Work performed on the gas: **$W_{comp,\ on} =$ +490.0 J** (Work done by gas $W = -490.0$ J).

> **Physical Sanity Check:** The compressed temperature of 953 K (680°C) vastly exceeds the diesel auto-ignition threshold (≈ 550 K / 277°C). This explains why diesel engines require no spark plugs: injected liquid fuel instantaneously self-ignites purely from the extreme thermal energy imparted by adiabatic compression work!

Problem 4: Carnot Refrigerator Cooling Power, Compressor Work & Operating Cost

Chapter 6 & 7 • Fast-Track TH-4

Problem Statement: A commercial Carnot deep-freezer maintains an internal food compartment at $T_C = -18.0°C$ (255.15 K) in a restaurant kitchen where the ambient temperature is $T_H = 27.0°C$ (300.15 K). Heat continuously infiltrates through the insulated walls and door gaskets at a steady rate of $H_{leak} = 850$ W. Calculate: (a) The theoretical maximum Coefficient of Performance ($COP_{Carnot,\ ref}$); (b) The minimum mechanical/electrical power P_{comp} in Watts required by the compressor motor to maintain steady temperature; (c) The rate of heat rejected into the kitchen $H_{kitchen}$; (d) The daily electrical operating cost if power is billed at \$0.160 per kilowatt-hour (kWh); (e) The required power and daily cost if a real-world compressor operates at only 45.0% of the theoretical Carnot limit.

Step 1: Given Parameters & Fast-Track Setup

$T_C = 273.15 - 18.0 = 255.15$ K, $T_H = 273.15 + 27.0 = 300.15$ K.

Temperature lift: $\Delta T = T_H - T_C = 300.15 - 255.15 = 45.00$ K.

Cooling rate required: $dQ_C/dt = H_{leak} = 850$ W.

Step 2: Symbolic Derivations

(a) $COP_{Carnot,\ ref} = T_C / \Delta T$.

(b) Since $COP = (dQ_C/dt) / P_{comp} \Rightarrow$ **$P_{comp} = (dQ_C/dt) / COP$**.

(c) From energy conservation: $dQ_H/dt = dQ_C/dt + P_{comp}$.

(d) Daily energy: $E_{kWh} = (P_{comp}$ [kW]) × 24 h. Daily cost = $E_{kWh} \times$ Rate.

Step 3: Numerical Evaluations

(a) **$COP_{Carnot,\ ref} = 255.15\ K / 45.00\ K = 5.670$** (Using Fast-Track TH-4).

(b) Minimum compressor electrical power:

$$P_{comp,\ Carnot} = 850\ W / 5.670 = 149.91\ W \approx 150\ W$$

(c) Rate of heat discharged into kitchen:

$H_{kitchen} = 850\ W + 149.91\ W = 999.91\ W \approx 1.00\ kW$.

(d) Daily cost for ideal Carnot unit:

$E_{day} = 0.14991$ kW × 24.0 h = 3.598 kWh/day.

Daily Cost = 3.598 kWh × \$0.160/kWh = \$0.576 / day (\$17.27 / month).

(e) Real compressor at 45.0% Carnot COP:

COP_{real} = 0.450 × 5.670 = **2.5515**.

$P_{\text{comp, real}}$ = 850 W / 2.5515 = **333.1 W**.

$E_{\text{real, day}}$ = 0.3331 kW × 24.0 h = 7.995 kWh/day.

Real Daily Cost = 7.995 × \$0.160 = \$1.28 / day (\$38.38 / month).

> **Physical Sanity Check:** A COP of 5.67 means that for every 1 Joule of electrical energy drawn by the compressor, 5.67 Joules of heat are successfully extracted from the freezer! Notice also that $H_{\text{kitchen}} > H_{\text{leak}}$: the refrigerator acts as a net room heater with thermal output equal to heat removed plus compressor electrical dissipation.

Problem 5: Irreversible Heat Exchange & Entropy Generation of the Universe

Chapter 8 • Fast-Track TH-5

Problem Statement: A solid block of pure copper of mass m_{Cu} = 0.600 kg initially at temperature $T_{\text{Cu, i}}$ = 100.0°C (373.15 K) is dropped into an insulated calorimeter containing m_{w} = 0.800 kg of liquid water at $T_{\text{w, i}}$ = 15.0°C (288.15 K). The container has negligible heat capacity. Specific heat capacities: c_{Cu} = 386 J/(kg·K), c_{w} = 4,186 J/(kg·K). Determine: (a) The final equilibrium temperature T_{f} of the mixture; (b) The entropy change of the copper block ΔS_{Cu}; (c) The entropy change of the water ΔS_{w}; (d) The net entropy change of the universe $\Delta S_{\text{univ}} = \Delta S_{\text{Cu}} + \Delta S_{\text{w}}$, proving that $\Delta S_{\text{univ}} > 0$ strictly in accordance with the Second Law of Thermodynamics.

Step 1: Final Equilibrium Temperature via Calorimetry Conservation

Heat lost by copper equals heat gained by water: $|Q_{\text{lost}}| = Q_{\text{gained}}$.

$m_{\text{Cu}}\, c_{\text{Cu}}\, (T_{\text{Cu, i}} - T_{\text{f}}) = m_{\text{w}}\, c_{\text{w}}\, (T_{\text{f}} - T_{\text{w, i}})$.

Thermal heat capacities:

$C_{\text{Cu}} = m_{\text{Cu}}\, c_{\text{Cu}}$ = 0.600 × 386 = **231.6 J/K**.

$C_{\text{w}} = m_{\text{w}}\, c_{\text{w}}$ = 0.800 × 4186 = **3,348.8 J/K**.

Solving for T_{f}:

$$T_{\text{f}} = \frac{C_{\text{Cu}}\, T_{\text{Cu, i}} + C_{\text{w}}\, T_{\text{w, i}}}{C_{\text{Cu}} + C_{\text{w}}} = \frac{(231.6 \times 373.15) + (3348.8 \times 288.15)}{231.6 + 3348.8} = \frac{86{,}421.5 + 964{,}957.7}{3{,}580.4} = 293.65\ \text{K}\ (20.50°\text{C})$$

Temperature changes: ΔT_{Cu} = 293.65 − 373.15 = −79.50 K; ΔT_{w} = 293.65 − 288.15 = +5.50 K.

Step 2: Entropy Calculations via Reversible Equivalent Paths

Because entropy is a state function, evaluate $\Delta S = \int (m\, c / T)\, dT = m\, c \ln(T_{\text{f}} / T_{\text{i}})$:

• **Entropy Change of Copper:**

$$\Delta S_{\text{Cu}} = C_{\text{Cu}} \ln(T_{\text{f}} / T_{\text{Cu, i}}) = 231.6 \times \ln(293.65 / 373.15) = 231.6 \times (-0.23961) = -55.49\ \text{J/K}$$

• **Entropy Change of Water:**

$$\Delta S_w = C_w \ln(T_f / T_{w,i}) = 3348.8 \times \ln(293.65 / 288.15) = 3348.8 \times (+0.018906) = +63.31 \text{ J/K}$$

Step 3: Total Entropy of the Universe & Second Law Attestation

The insulated container exchanges no heat with the outside environment, so $\Delta S_{surroundings} = 0$.

The total entropy generation of the universe is:

$$\Delta S_{universe} = \Delta S_{Cu} + \Delta S_w = -55.49 \text{ J/K} + 63.31 \text{ J/K} = +7.82 \text{ J/K} > 0$$

Physical Sanity Check: Although the copper block experienced an entropy decrease (−55.49 J/K) as heat was extracted from it, the water absorbed that same thermal energy at a lower average temperature, resulting in a larger entropy gain (+63.31 J/K). The net entropy generation is strictly positive ($\Delta S_{univ} = +7.82$ J/K > 0), rigorously validating the Clausius Principle of Entropy Increase for spontaneous irreversible thermal equalization!

Part V: Reference Appendices & Master Summary Cheat Sheet

Appendix A: Master Comparison Matrix of Ideal Gas Thermodynamic Processes

Process Type	Governing Condition	Work Done (W)	Heat Added (Q)	Internal Energy (ΔE_{int})	Entropy Change (ΔS)
Isobaric	P = constant V / T = const	$P \Delta V = n R \Delta T$	$n C_p \Delta T$	$n C_v \Delta T$	$n C_p \ln(T_f / T_i)$
Isochoric	V = constant P / T = const	0	$n C_v \Delta T$	$n C_v \Delta T$	$n C_v \ln(T_f / T_i)$
Isothermal	T = constant $P V$ = const	$n R T \ln(V_f / V_i)$	W	0	$n R \ln(V_f / V_i) = Q/T$
Adiabatic	$Q = 0$ $P V^{\gamma}$ = const	$(P_i V_i - P_f V_f)/(\gamma-1)$	0	$-W = n C_v \Delta T$	0 (Reversible isentropic)
Free Expansion	Insulated vacuum $Q = 0$, $W = 0$	0	0	0 ($\Delta T = 0$)	$n R \ln(V_f / V_i) > 0$

Appendix B: Heat Transfer Mechanisms Master Formula Table

Mechanism	Governing Physical Law	Primary Formula	Key Physical Parameters	Thermal Resistance Analog
Conduction	Fourier's Law	$H = k A (T_H - T_C) / L$	k: Thermal conductivity [W/(m·K)] A: Area, L: Thickness	$R_{th} = L / (k A)$ Series: $\sum R_i$
Convection	Newton's Law of Cooling	$H = h A (T_s - T_\infty)$	h: Convective coefficient [W/(m^2·K)] T_s: Surface temp, T_∞: Fluid temp	$R_{conv} = 1 / (h A)$
Radiation	Stefan-Boltzmann Law	$P_{net} = \varepsilon \sigma A (T^4 - T_{env}^4)$	$\sigma = 5.670 \times 10^{-8}$ W/(m^2·K^4) ε: Surface emissivity ($0 \le \varepsilon \le 1$)	Non-linear radiation resistance
Wien Peak	Wien's Displacement Law	$\lambda_{max} T = 2.898 \times 10^{-3}$ m·K	Peak spectral emission wavelength	Spectral distribution marker

Appendix C: Master Thermodynamics & Heat Transfer Cheat Sheet

Concept / Phenomenon	Standard Physical Equation	⚡ Fast-Track Hand Calculation Shortcut
Linear & Volume Expansion	$\Delta L = \alpha L_0 \Delta T, \quad \Delta V = \beta V_0 \Delta T$	Use 1-2-3 rule: $\beta = 3\alpha$, $\gamma = 2\alpha$. Stress: $\sigma = E \alpha \Delta T$
Latent Heat Phase Change	$Q = \pm m L_f, \quad Q = \pm m L_v$	Water: $L_f \approx 333$ kJ/kg, $L_v \approx 2260$ kJ/kg $\approx 6.8 \times L_f$
Series Conduction Interface	$k_1 A(T_H - T_{int})/L_1 = k_2 A(T_{int} - T_C)/L_2$	Weighted average: $T_{int} = (G_1 T_H + G_2 T_C)/(G_1 + G_2)$
Isobaric Work Fraction	$W = P \Delta V, \quad Q = n C_p \Delta T$	Monatomic: $W = 0.40Q$. Diatomic: $W = (2/7)Q \approx 0.286Q$
Adiabatic Compression Temp	$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$	$T_2 = T_1 \cdot r^{\gamma-1}$. For air ($\gamma = 1.4$): $T_2 = T_1 \cdot r^{0.4}$
Carnot Efficiency	$\eta_C = 1 - T_C / T_H$	$\eta_C = \Delta T / T_H$. Express as simple fraction
Carnot Refrigerator COP	$COP_{Carnot} = T_C / (T_H - T_C)$	COP $= (1 - \eta_C) / \eta_C$. If $\eta = 25\%$, COP $= 0.75/0.25 = 3$
Heat Pump vs Refrigerator	$COP_{hp} = Q_H / W, \quad COP_{ref} = \lvert Q_C \rvert / W$	Universal identity: **$COP_{hp} = COP_{ref} + 1$**
Ideal Gas Entropy Change	$\Delta S = n C_v \ln(T_f/T_i) + n R \ln(V_f/V_i)$	Isothermal volume doubling: $\Delta S = n R \ln 2 \approx 0.693\, n R$
Phase Change Entropy	$\Delta S = \int dQ_{rev} / T$	Isothermal transition: $\Delta S = \pm m L / T_{trans}$